



PHOTO FRAME GS 02

E-Ink Display 960x540px [LilyGo T5 E-paper Plus / ESP32 S3]

E-ink digital picture frame, featuring SD + LittleFS image storage with remote management via FTP/Telnet (including HTTP(S) fetch). Includes dual-slot OTA firmware updates and ultra-low-power deep sleep with RTC backup for long battery life.

CONNECTION & ACCESS

WIFI AP MODE (Default)

SSID: PhotoFrameGS02
Password: 123456789
IP: 192.168.4.1
Gateway: 192.168.4.1
Subnet: 255.255.255.0

WIFI STA MODE

sta_ssid – WiFi name
sta_password – WiFi password
sta_enable – static IP
sta_ip/gateway/subnet/dns
Set: ap_enable = false

mDNS (optional)

mdns_enable = true
mdns_hostname = photoframegs02
Access: photoframegs02.local

TELNET ACCESS

Port: 23
User: admin
Pass: 123456789
Session: 1 hour

FTP ACCESS

Port: 21
User: admin
Pass: 123456789

TELNET COMMANDS

SYSTEM

help – command list
clear – clear screen
reboot – restart
reset config – factory reset
exit – disconnect
logout – re-auth

INFORMATION

meminfo – RAM usage
netinfo – IP/MAC/DNS info
fsinfo – storage usage
sketchinfo – firmware/sketch info
nvinfo – NVS usage
batinfo – battery info
date – current time
date rtc – RTC time
date rtc set – set RTC time
date rtc sync-from-ntp – sync RTC from NTP
date rtc sync-to-system – sync time from RTC
timestamp – Unix epoch

FILES

list – show all files
cat <filename> – view content

CONFIGURATION

config <key> – get configuration
config <key> [value] – set configuration

DOWNLOAD

fetch <url> [name] – HTTP/HTTPS
– max 200KB, j(e)pg

FTP & CONFIGURATION

FILE STRUCTURE

/config.ini – settings
/images/ – photos

IMAGE REQUIREMENTS

Format: JPG/JPEG
Size: 960x540px
Max.: 200 KB per file
Tip: Grayscale for best

magic pic01.jpg --colorspace Gray --contrast-stretch 2000 --redshift-dither e0d0,16 pic01.jpg

CONFIG.INI SECTIONS

[device] – name, version
[display] – jpg settings
[ap mode] – AP settings
[sta mode] – WiFi creds
[static ip] – IP settings
[ntp] – time server
[telnet] – telnet config
[ftp] – ftp config
[timer] – wake interval

WAKE UP VALUES

1=Sec, **2**=Min, **3**=Hour,
4=12h, **5**=Day, **6**=Week, **7**=Month

BUTTONS

Next Image (1s), Reset: (Factory Reset 30s),
Wake/Maintenance (1s)

OPERATING MODES

1 PHOTO FRAME MODE (Normal Operation)

- Automatically displays JPG images from **/images/** folder
- Cycles through images based on configured wake interval
- Supports:** Seconds, Minutes, Hourly, Half-Day, Daily, Weekly, Monthly intervals
- Deep sleep between image changes for battery conservation
- Low battery detection with visual warning

2 MAINTENANCE MODE (Configuration & File Management)

Activation: Hold Maintenance Button (GPIO 21) for 3 seconds during boot

Features available:

- WiFi connection (AP or STA mode)
- Telnet server for remote configuration
- FTP server for file upload/download
- mDNS support for easy hostname access

Exit: Hold Maintenance Button for 3 seconds to reboot

Factory Reset: Hold Reset Button (GPIO 48) for 30 seconds

3 LOW BATTERY MODE (Empty battery)

- Triggered when battery voltage is too low
- Displays battery icon with voltage and percentage
- Enters extended deep sleep to preserve battery

DESIGNED BY



FROM



BUTTONS

NEXT IMAGE

RESET

MAIN-TENANCE

USB-C CONNECTOR